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(54) **BEARING ASSEMBLY WITH LOAD ISOLATION FOR GAS TURBINE ENGINE**

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F16C 35/077 (2006.01)

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CPC **F16C 19/522** (2013.01); **F01D 25/162** (2013.01); **F16C 35/077** (2013.01); **F05D 2240/50** (2013.01); **F16C 2360/23** (2013.01)

(58) **Field of Classification Search**

CPC **F16C 19/522**; **F16C 35/042**; **F16C 35/045**; **F16C 35/067**; **F16C 35/077**; **F16C 2360/23**; **F01D 25/162**; **F05D 2240/50**

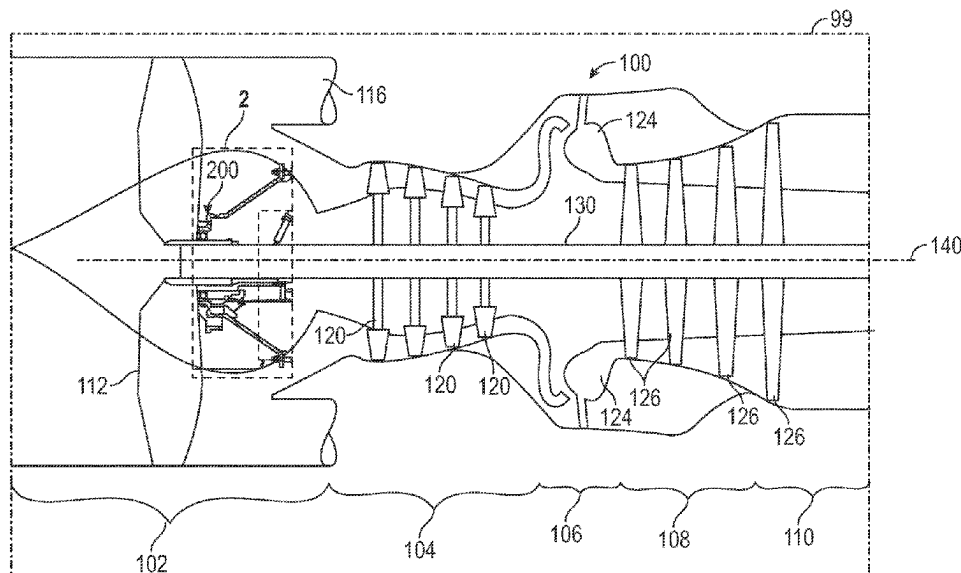
See application file for complete search history.

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ABSTRACT

A bearing assembly includes an inner race to receive a rotating component having an axis of rotation, and at least one bearing element coupled to the inner race. The at least one bearing element is to rotate with the inner race along the axis of rotation. The bearing assembly includes an outer race coupled to the at least one bearing element, and a bearing housing coupled to the outer race and to be coupled to a static structure. The bearing housing includes a first frangible portion to release upon receipt of a moment load greater than a moment threshold and a second frangible portion to release upon receipt of a radial load greater than a radial threshold.

20 Claims, 9 Drawing Sheets



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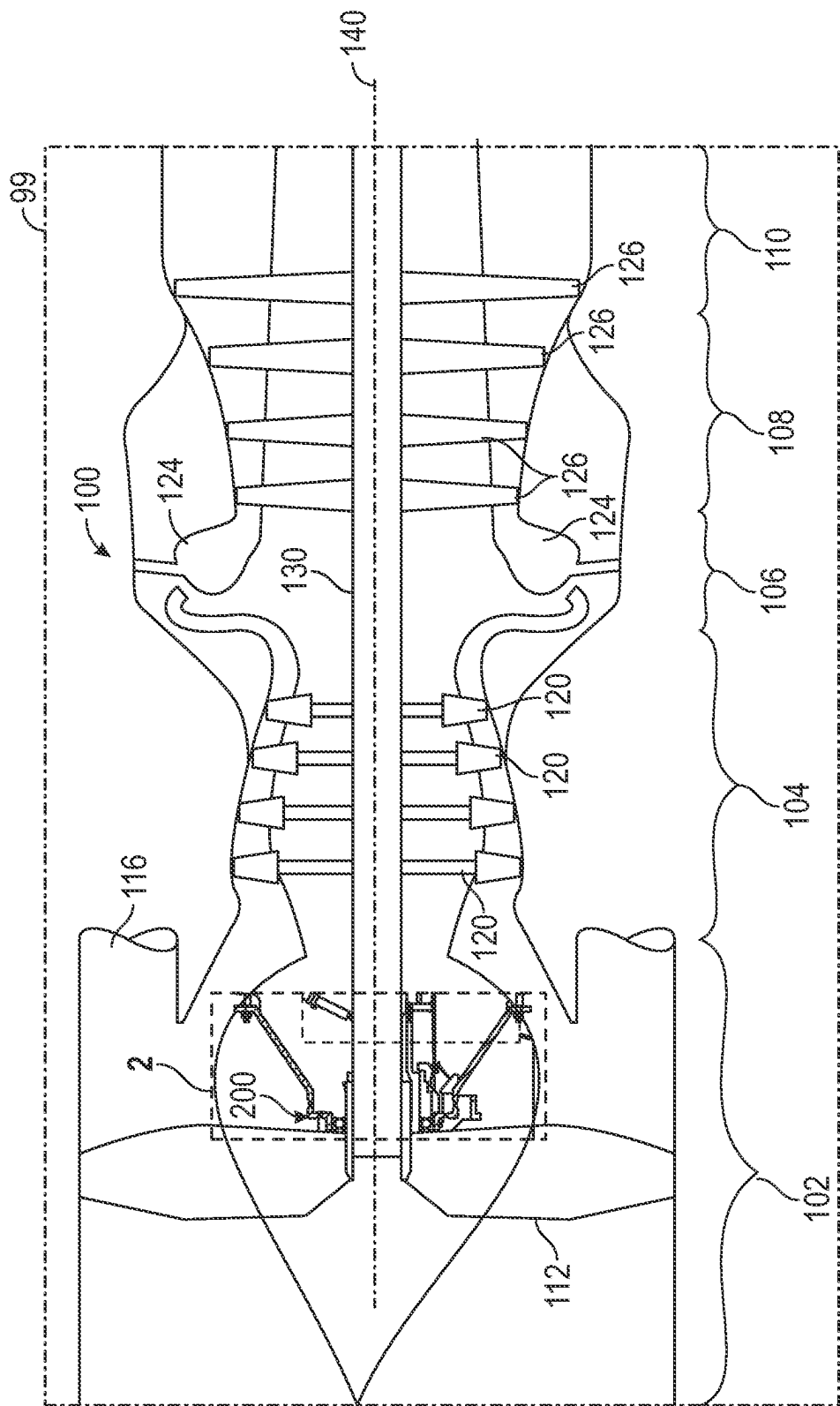


FIG. 1

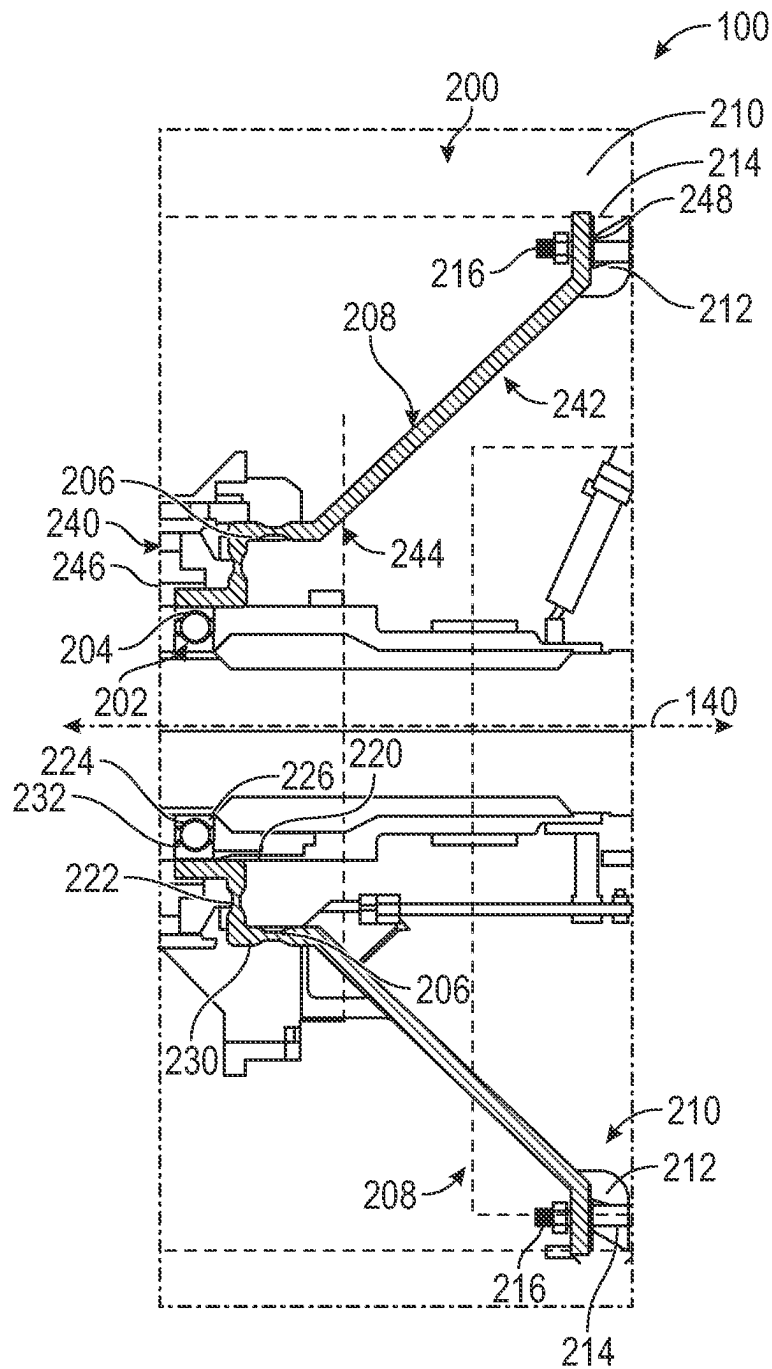


FIG. 2

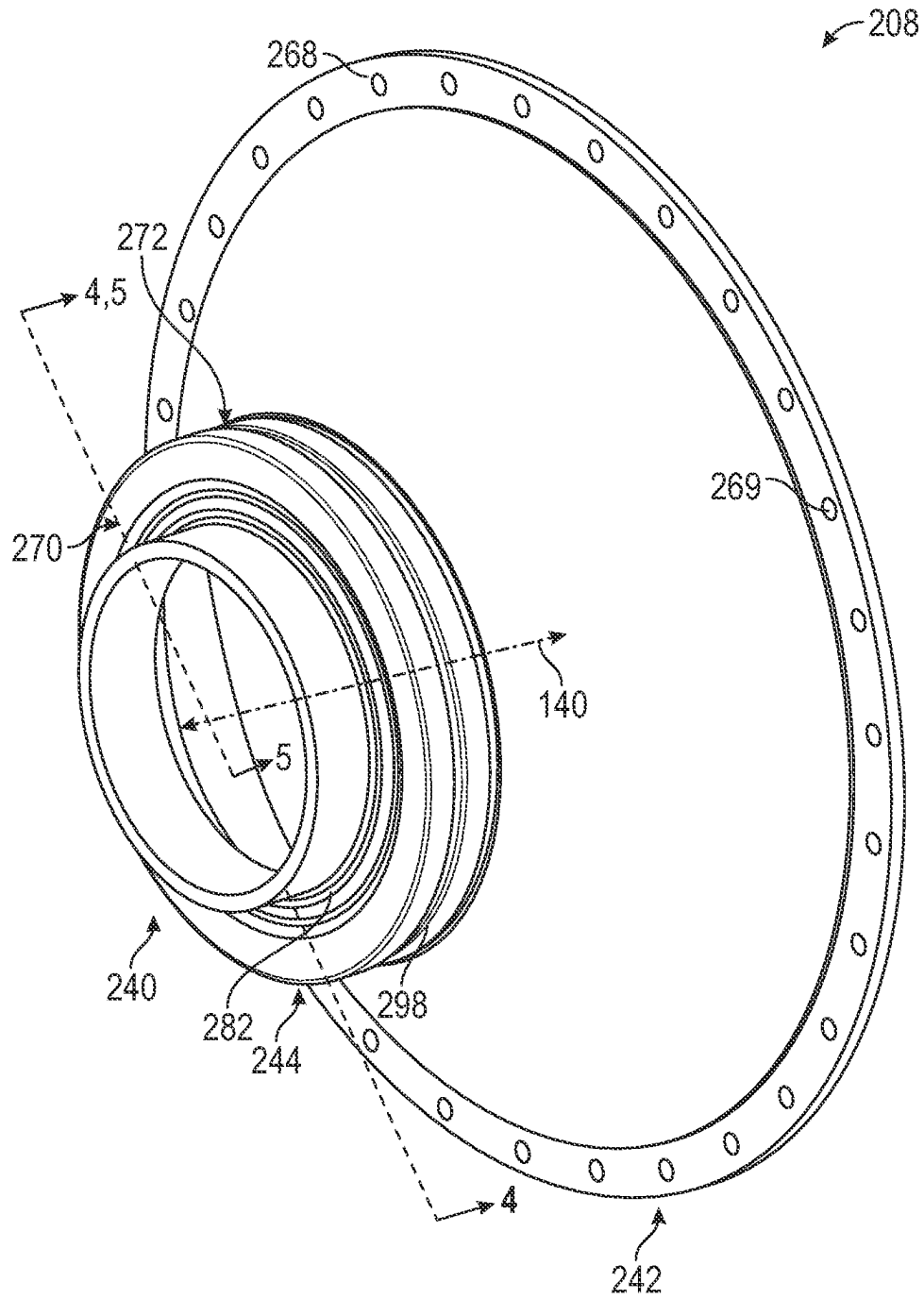


FIG. 3

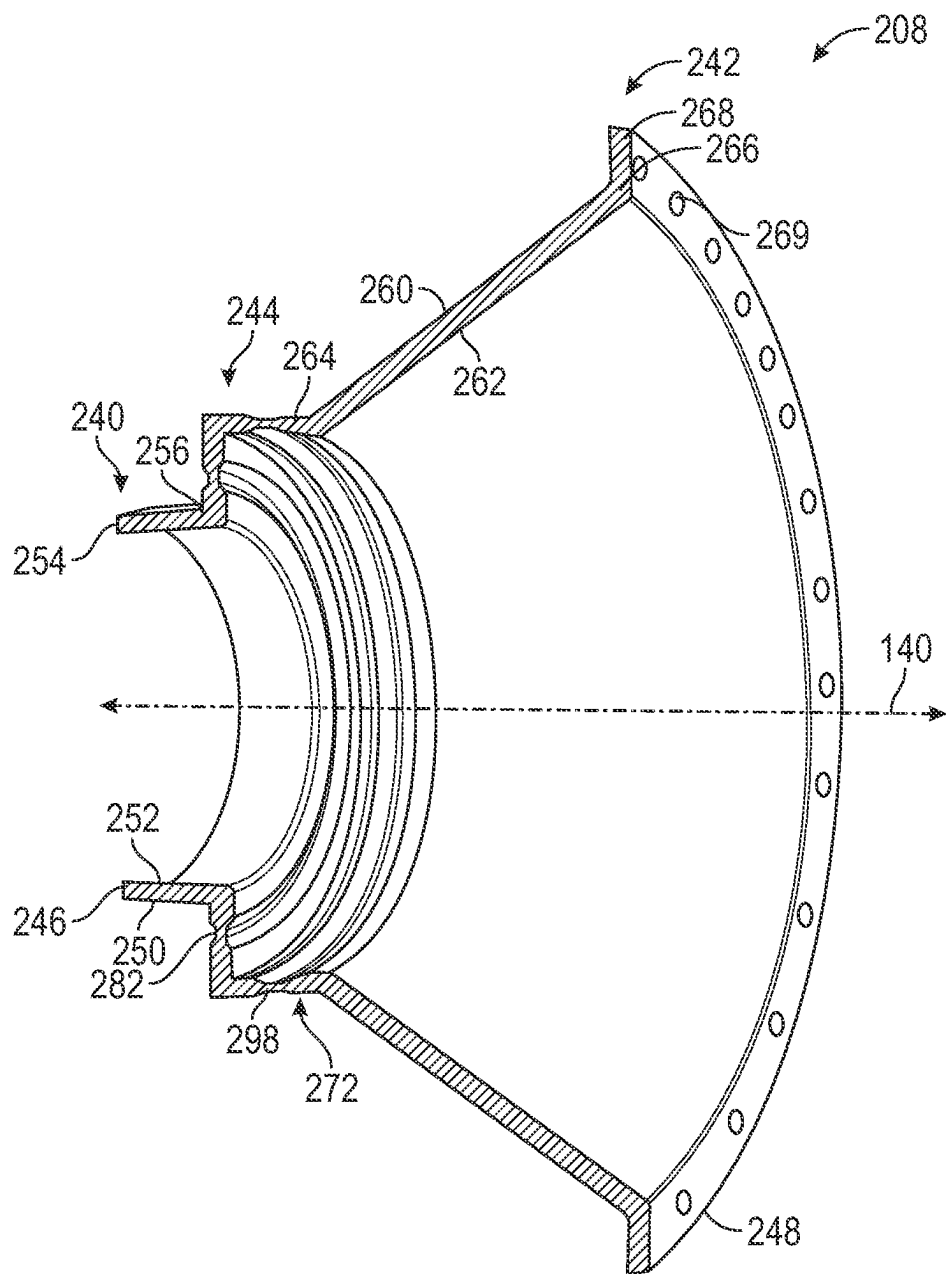


FIG. 4

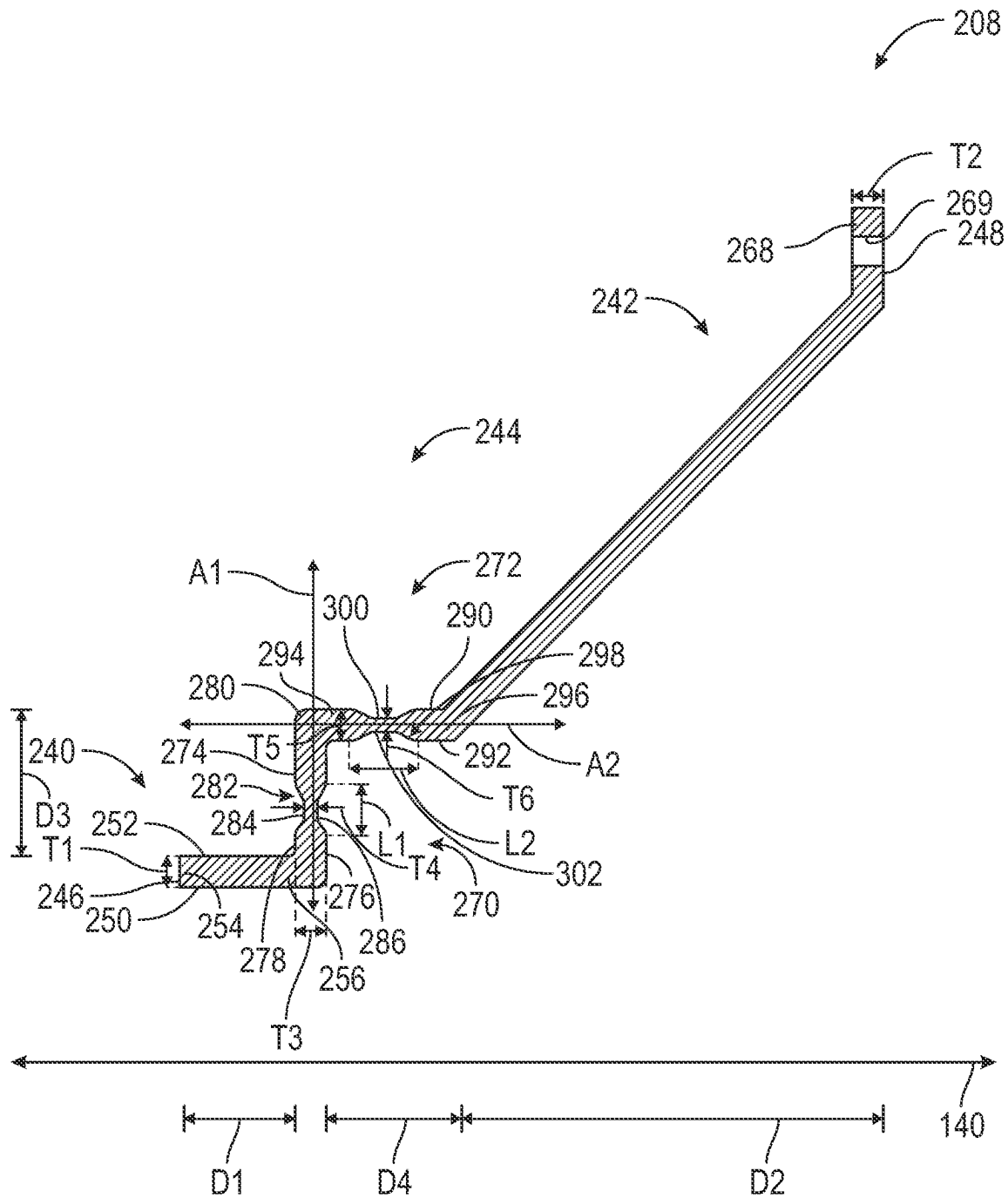


FIG. 5

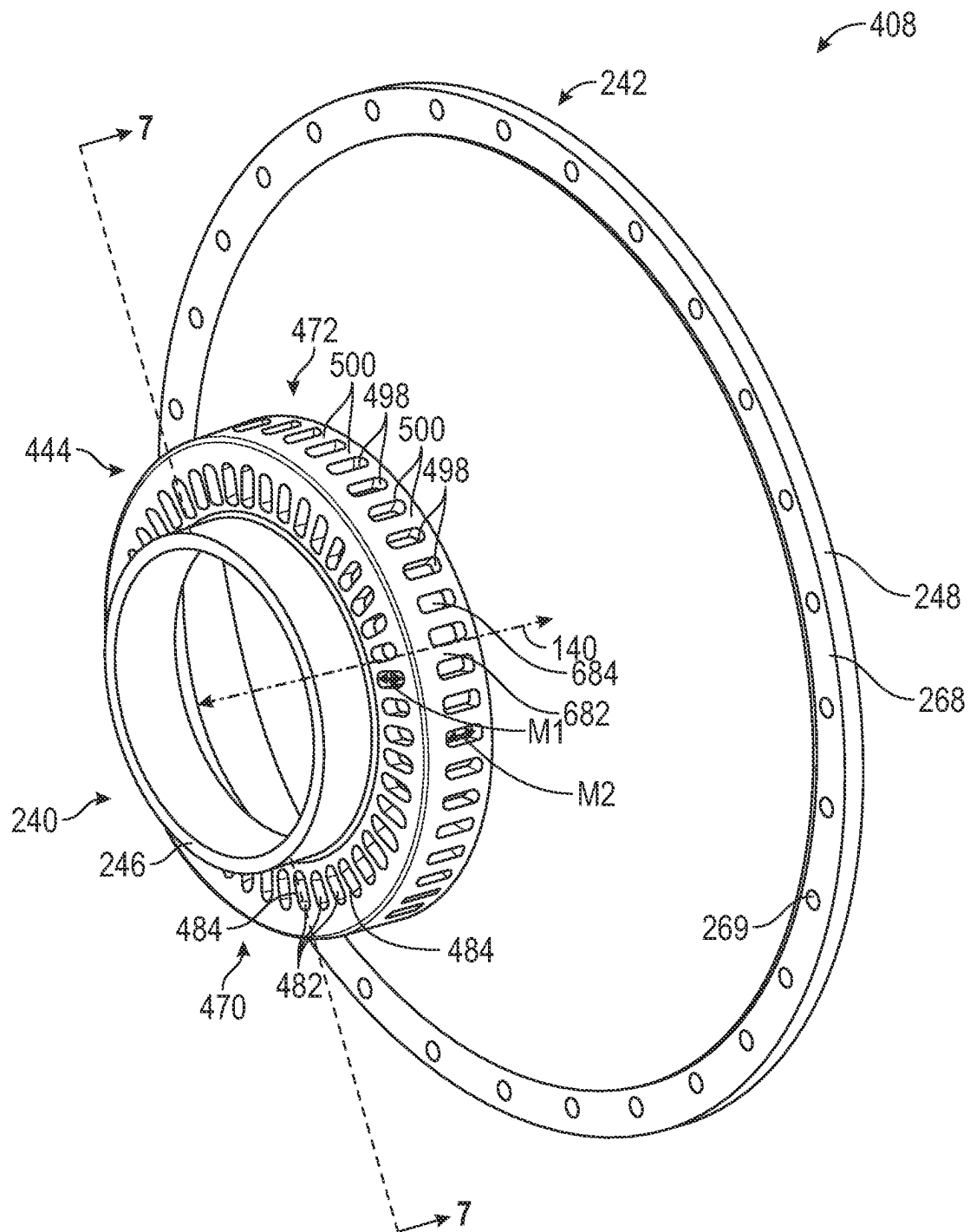


FIG. 6

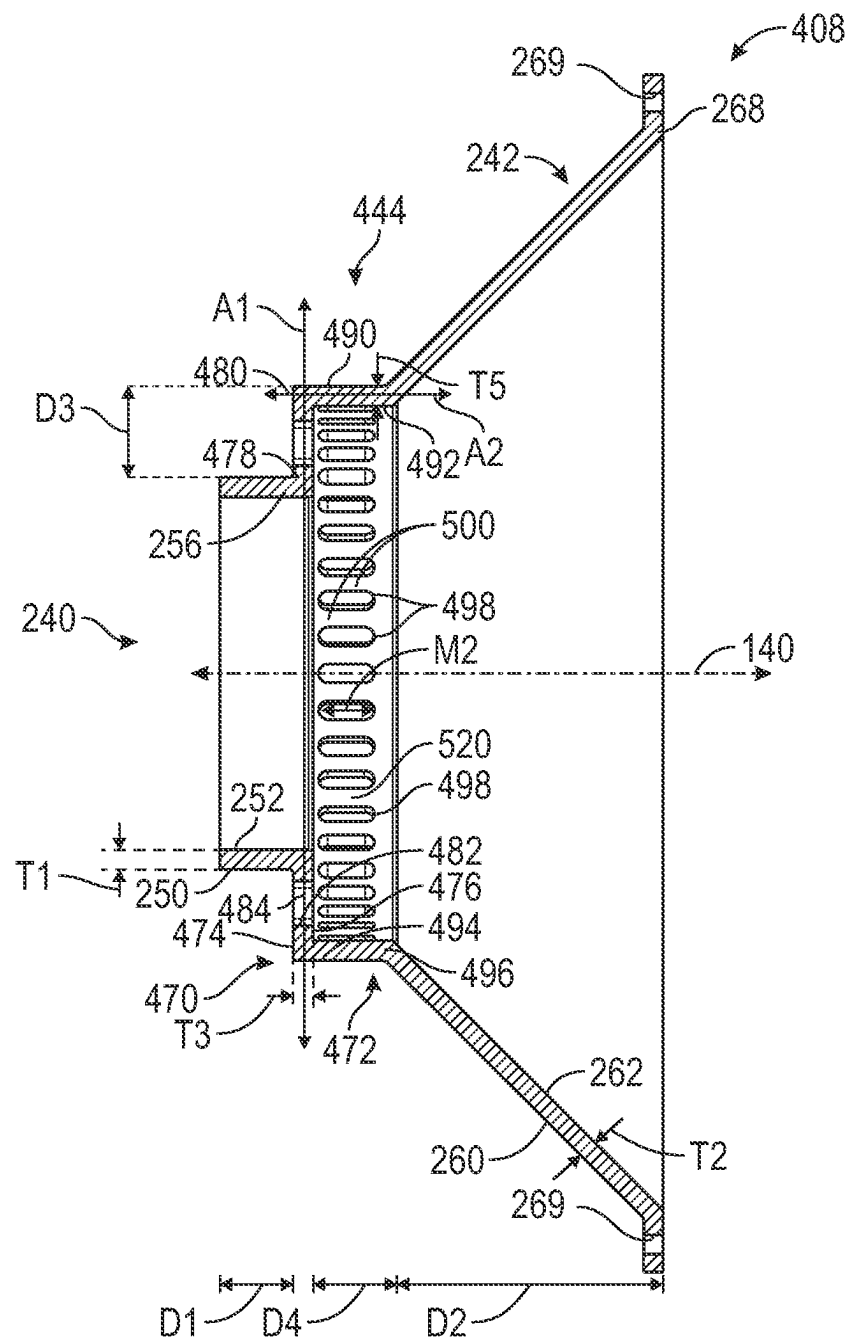


FIG. 7

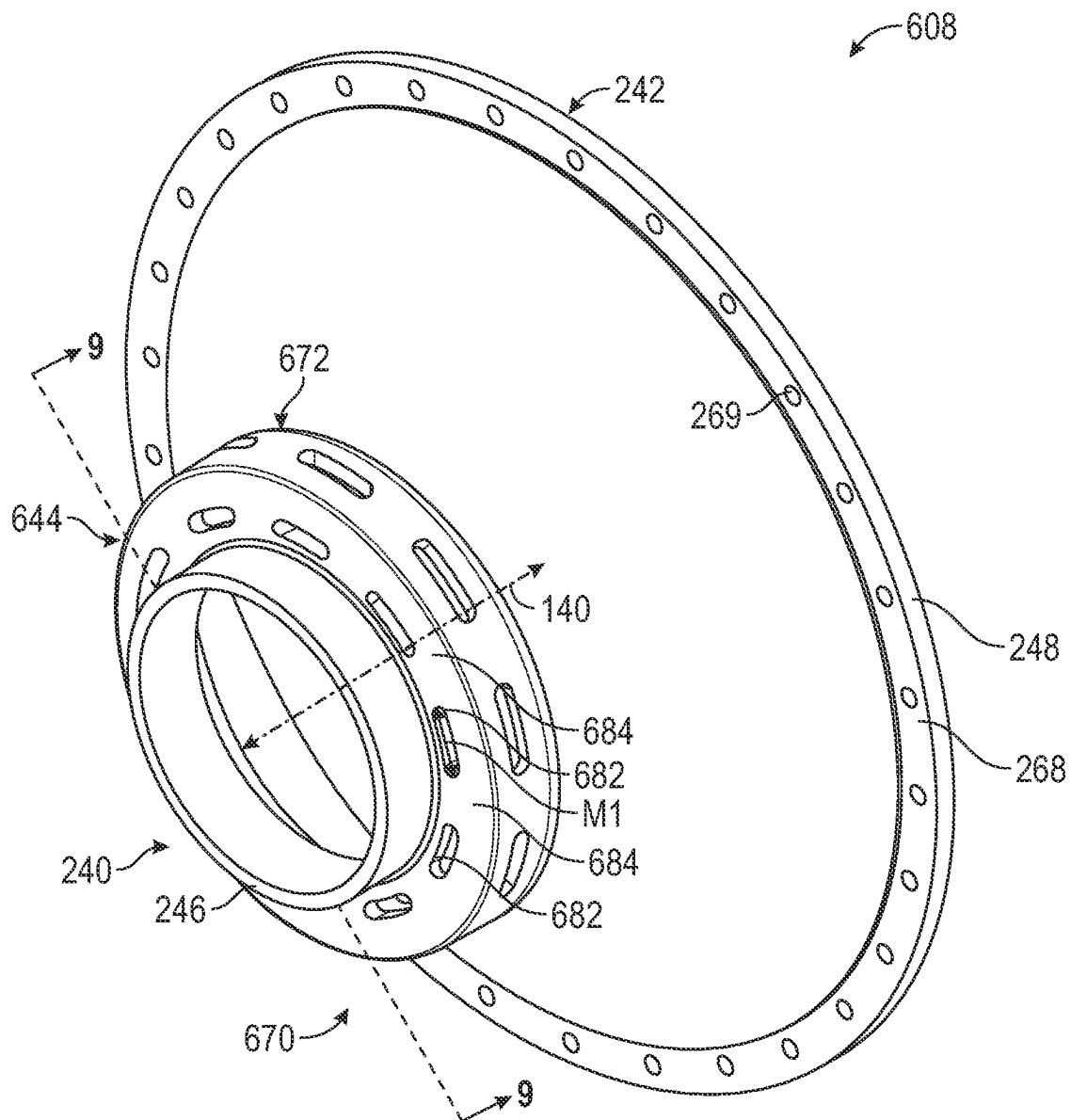


FIG. 8

FIG. 9

1

BEARING ASSEMBLY WITH LOAD ISOLATION FOR GAS TURBINE ENGINE

CROSS REFERENCE TO RELATED APPLICATION

This application claims priority to Indian Provisional Patent Application No. 202311016104, filed Mar. 10, 2023, the entire content of which is incorporated by reference herein.

TECHNICAL FIELD

The present disclosure generally relates to gas turbine engines, and more particularly relates to a bearing assembly with load isolation for a gas turbine engine.

BACKGROUND

Gas turbine engines may be employed to power various devices. For example, a gas turbine engine may be employed to power a mobile platform, such as an aircraft or rotorcraft. Gas turbine engines may include one or more rotating components, which are supported by one or more bearings. Generally, a portion of the bearing is coupled to a static structure of the gas turbine engine. In certain instances, the rotating component may introduce loads into the bearing, which may be transferred to the static structure through the bearing. The transfer of the loads from the bearing to the static structure is undesirable as it may impact the performance of the static structure.

Accordingly, it is desirable to provide a bearing assembly with load isolation for a gas turbine engine, which inhibits the transfer of loads to a static structure associated with the gas turbine engine through the bearing assembly. Furthermore, other desirable features and characteristics of the present disclosure will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing technical field and background.

SUMMARY

According to various embodiments, provided is a bearing assembly. The bearing assembly includes an inner race configured to receive a rotating component having an axis of rotation, and at least one bearing element coupled to the inner race. The at least one bearing element is configured to rotate with the inner race along the axis of rotation. The bearing assembly includes an outer race coupled to the at least one bearing element, and a bearing housing coupled to the outer race and configured to be coupled to a static structure. The bearing housing includes a first frangible portion configured to release upon receipt of a moment load greater than a moment threshold and a second frangible portion configured to release upon receipt of a radial load greater than a radial threshold.

The first frangible portion is defined on the bearing housing along an axis substantially perpendicular to the axis of rotation. The second frangible portion is defined on the bearing housing along an axis substantially parallel to the axis of rotation. The first frangible portion includes at least one first groove that defines a reduced thickness. The second frangible portion includes at least one second groove that defines a second reduced thickness. The at least one first groove is continuous about a perimeter of the first frangible portion. The at least one first groove comprises a pair of

2

opposed grooves that cooperate to define the reduced thickness. The first frangible portion includes a plurality of first slots spaced apart about a perimeter of the first frangible portion. The second frangible portion includes a plurality of second slots spaced apart about a second perimeter of the second frangible portion. Each second slot extends along a major axis, and the major axis is substantially parallel to the axis of rotation. Each first slot extends along a major axis, and the major axis is substantially perpendicular to the axis of rotation. The bearing housing includes a first coupling section to be coupled to the outer race, a second coupling section configured to be coupled to the static structure, and the first frangible portion and the second frangible portion are defined between the first coupling section and the second coupling section. The first frangible portion extends radially outward from the first coupling section, and the second frangible portion extends axially from the first frangible portion relative to the axis of rotation. The second frangible portion is coupled to a first end of the second coupling section, and the second coupling section defines a flange at a second end, with the second end opposite the first end. The rotating component is a shaft associated with a gas turbine engine.

Further provided is a bearing assembly for a gas turbine engine. The bearing assembly includes an inner race configured to receive a rotating component having an axis of rotation. The bearing assembly includes at least one bearing element coupled to the inner race, and the at least one bearing element is configured to rotate with the inner race along the axis of rotation. The bearing assembly includes an outer race coupled to the at least one bearing element, and a bearing housing coupled to the outer race and configured to be coupled to a static structure. The bearing housing includes a first frangible portion and a second frangible portion. The first frangible portion extends along a first axis substantially perpendicular to the axis of rotation, and the first frangible portion is configured to release upon receipt of a moment load greater than a moment threshold. The second frangible portion extends along a second axis substantially parallel to the axis of rotation and the second frangible portion is configured to release upon receipt of a radial load greater than a radial threshold.

The first frangible portion includes a pair of opposed grooves that cooperate to define a reduced thickness and each groove of the pair of opposed grooves is continuous about a perimeter of the first frangible portion. The first frangible portion includes a plurality of first slots spaced apart about a perimeter of the first frangible portion and the second frangible portion includes a plurality of second slots spaced apart about a second perimeter of the second frangible portion. Each second slot extends along a major axis, the major axis is substantially parallel to the axis of rotation. Each first slot extends along a major axis, and the major axis is substantially perpendicular to the axis of rotation.

DESCRIPTION OF THE DRAWINGS

The exemplary embodiments will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:

FIG. 1 is a schematic cross-sectional illustration of a gas turbine engine, which includes an exemplary bearing assembly with load isolation in accordance with the various teachings of the present disclosure;

FIG. 2 is a detail cross-sectional view, taken at 2 on FIG. 1, of the bearing assembly with load isolation coupled about

3

a shaft associated with the gas turbine engine in accordance with the various teachings of the present disclosure;

FIG. 3 is a perspective view of an exemplary bearing housing associated with the bearing assembly;

FIG. 4 is a cross-sectional view of the bearing housing of FIG. 3 taken along line 4-4 of FIG. 3;

FIG. 5 is a cross-sectional view of the bearing housing of FIG. 3 taken along line 5-5 of FIG. 3;

FIG. 6 is another exemplary bearing housing for use with the bearing assembly of FIG. 1;

FIG. 7 is a cross-sectional view of the bearing housing of FIG. 6 taken along line 7-7 of FIG. 6;

FIG. 8 is yet another exemplary bearing housing for use with the bearing assembly of FIG. 1; and

FIG. 9 is a cross-sectional view of the bearing housing of FIG. 8 taken along line 9-9 of FIG. 8.

DETAILED DESCRIPTION

The following detailed description is merely exemplary in nature and is not intended to limit the application and uses. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with any type of engine that would benefit from a bearing assembly with load isolation and the gas turbine engine described herein is merely one exemplary embodiment according to the present disclosure. In addition, while the bearing assembly is described herein as being used with a gas turbine engine onboard a mobile platform, such as a bus, motorcycle, train, motor vehicle, marine vessel, aircraft, rotorcraft and the like, the various teachings of the present disclosure can be used with a gas turbine engine on a stationary platform. Further, it should be noted that many alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure. In addition, while the figures shown herein depict an example with certain arrangements of elements, additional intervening elements, devices, features, or components may be present in an actual embodiment. It should also be understood that the drawings are merely illustrative and may not be drawn to scale.

As used herein, the term “axial” refers to a direction that is generally parallel to or coincident with an axis of rotation, axis of symmetry, or centerline of a component or components. For example, in a cylinder or disc with a centerline and generally circular ends or opposing faces, the “axial” direction may refer to the direction that generally extends in parallel to the centerline between the opposite ends or faces. In certain instances, the term “axial” may be utilized with respect to components that are not cylindrical (or otherwise radially symmetric). For example, the “axial” direction for a rectangular housing containing a rotating shaft may be viewed as a direction that is generally parallel to or coincident with the rotational axis of the shaft. Furthermore, the term “radially” as used herein may refer to a direction or a relationship of components with respect to a line extending outward from a shared centerline, axis, or similar reference, for example in a plane of a cylinder or disc that is perpendicular to the centerline or axis. In certain instances, components may be viewed as “radially” aligned even though one or both of the components may not be cylindrical (or otherwise radially symmetric). Furthermore, the terms “axial” and “radial” (and any derivatives) may encompass directional relationships that are other than precisely aligned

4

with (e.g., oblique to) the true axial and radial dimensions, provided the relationship is predominantly in the respective nominal axial or radial direction. As used herein, the term “about” denotes within 10% to account for manufacturing tolerances. In addition, the term “substantially” denotes within 10% to account for manufacturing tolerances.

With reference to FIG. 1, a simplified cross-sectional view of an exemplary gas turbine engine 100 is shown with the remaining portion of the gas turbine engine 100 being substantially axisymmetric about an axis of rotation 140, which also comprises a longitudinal axis for the gas turbine engine 100. As will be discussed herein, the gas turbine engine 100 includes a bearing assembly 200 with load isolation, which inhibits loads, such as shear loads and moment loads, from passing through the bearing assembly 200 into a surrounding fixed or static structure. In the depicted embodiment, the gas turbine engine 100 is an annular multi-spool turbofan gas turbine jet engine for use with a vehicle, such as an aircraft 99, although other arrangements and uses may be provided. For example, in other embodiments, the gas turbine engine 100 may assume the form of a non-propulsive engine, such as an Auxiliary Power Unit (APU) deployed onboard the aircraft 99, or an industrial power generator.

In this example, the gas turbine engine 100 includes a fan section 102, a compressor section 104, a combustor section 106, a turbine section 108, and an exhaust section 110. The fan section 102 includes a fan 112 that draws air into the gas turbine engine 100 and accelerates it. A fraction of the accelerated air exhausted from the fan 112 is directed through an outer (or first) bypass duct 116 and the remaining fraction of air exhausted from the fan 112 is directed into the compressor section 104. In the embodiment of FIG. 1, the compressor section 104 includes one or more compressors 120, and in one example, includes four compressors 120. However, in other embodiments, the number of compressors 120 in the compressor section 104 may vary. In the depicted embodiment, the compressors 120 sequentially raise the pressure of the air and direct a majority of the high pressure air into the combustor section 106. A fraction of the compressed air bypasses the combustor section 106 and is used to cool, among other components, turbine blades in the turbine section 108.

In the embodiment of FIG. 1, in the combustor section 106, which includes a combustion chamber 124, the high pressure air is mixed with fuel, which is combusted. The high-temperature combustion air is directed into the turbine section 108. In this example, the turbine section 108 includes one or more turbines 126, and in one example, includes four turbines 126 disposed in axial flow series. However, it will be appreciated that the number of turbines, and/or the configurations thereof, may vary. In this embodiment, the high-temperature air from the combustor section 106 expands through and rotates each turbine 126. As the turbines 126 rotate, each drives equipment in the gas turbine engine 100 via at least one concentrically disposed spool or shaft 130. The rotation of the spool or shaft 130 provides power output, which may be utilized in a variety of different manners, depending upon whether the gas turbine engine 100 assumes the form of a turbofan, turboprop, turboshaft, turbojet engine, or an auxiliary power unit, to list but a few examples. The spool or shaft 130 rotates about the axis of rotation 140.

In this example, the bearing assembly 200 is coupled to a rotating component having an axis of rotation, such as the shaft 130 having the axis of rotation 140. The bearing assembly 200 is coupled to the shaft 130 proximate, down-

5

stream of or behind the fan **112** in the fan section **102**. The bearing assembly **200** supports the shaft **130** for rotation. It should be noted that in other embodiments, the bearing assembly **200** may be used to support rotation of other rotating components associated with the gas turbine engine **100** and the location of the bearing assembly **200** is merely an example. With reference to FIG. 2, a detail cross-sectional view of a portion of the fan section **102** is shown. In one example, the bearing assembly **200** includes an inner race **202**, one or more bearing elements **204**, an outer race **206** and a bearing housing **208**. In this example, the inner race **202** is coupled to the shaft **130**, and the bearing housing **208** is coupled to a static structure **210**, which in one example, is a flange **212** that is coupled to a strut **214** associated with the compressor section **104** (FIG. 1). The bearing housing **208** is coupled to the flange **212** via one or more mechanical fasteners, such as bolts **216**. It should be noted that while not illustrated herein, the bearing assembly **200** may also include one or more seals.

As discussed, the inner race **202** is coupled to the shaft **130** via a press-fit, for example, but swaging and other techniques may be used to couple the inner race **202** to the shaft **130**. The inner race **202** is coupled of a metal or metal alloy, and is cast, forged, stamped, additively manufactured, etc. The inner race **202** is annular, and is sized to be coupled about the diameter of the shaft **130** to support the shaft **130** for rotation relative to the outer race **206** and the static structure **210**. The inner race **202** has a first inner surface **220** opposite a second inner surface **222**. The first inner surface **220** is coupled to the shaft **130**, and the second inner surface **222** is coupled to the bearing elements **204**. In one example, the second inner surface **222** may include a concave inner recess **224**, which is shaped to receive and partially retain the bearing elements **204**. The second inner surface **222** defines an inner perimeter or diameter of the inner race **202**, and surrounds a bore defined through the inner race **202**.

The bearing elements **204** enable the rotation of the inner race **202** relative to the outer race **206**. In one example, the bearing elements **204** comprise a plurality of ball bearings. In other examples, the bearing elements **204** may comprise a plurality of roller bearings, cylindrical roller bearings, needle bearings, etc. The bearing elements **204** are composed of a metal or metal alloy, and may be machined, additively manufactured, etc. The bearing elements **204** may be arranged between the inner race **202** and the outer race **206**, and retained by the inner recess **224** and a concave outer recess **226** of the outer race **206**. In other embodiments, the bearing elements **204** may be supported in a cage.

The outer race **206** is coupled to the bearing housing **208** via a press-fit, for example, but other techniques may be used, such as welding, mechanical fasteners, etc. The outer race **206** is coupled of a metal or metal alloy, and is cast, forged, stamped, additively manufactured, etc. The outer race **206** is annular, and is sized to be coupled about the bearing elements **204** and the inner race **202**. The outer race **206** has a first outer surface **230** opposite a second outer surface **232**. The first outer surface **230** is coupled to the bearing housing **208**, and the second outer surface **232** is coupled to the bearing elements **204**. In one example, the second outer surface **232** includes the outer recess **226**, which is shaped to receive and partially retain the bearing elements **204**. The bearing elements **204** rotate relative to the outer race **206**. The second outer surface **232** defines an inner perimeter or diameter of the outer race **206**, and surrounds a bore defined through the outer race **206**.

The bearing housing **208** is coupled to the outer race **206**, and to the flange **212** associated with the static structure **210**.

6

The bearing housing **208** is composed of a metal or metal alloy, and is cast, forged, stamped, machined, additively manufactured, etc. The bearing housing **208** includes a first coupling section or race coupling section **240**, a second coupling section or static coupling section **242** and a frangible section **244** between the race coupling section **240** and the static coupling section **242**. In this example, the bearing housing **208** is substantially symmetric about the axis of rotation **140**. The race coupling section **240** is at a first housing end **246**, and the static coupling section **242** extends from the frangible section **244** to a second housing end **248**. The second housing end **248** is opposite the first housing end **246**.

With reference to FIG. 3, the bearing housing **208** is shown in greater detail. The race coupling section **240** is cylindrical, and is sized to be positioned about the outer race **206**, the bearing elements **204** and the inner race **202**. With reference to FIG. 4, the race coupling section **240** includes an outer race surface **250** opposite an inner race surface **252**. The outer race surface **250** forms an outer surface or outer perimeter of the race coupling section **240**, and the inner race surface **252** is coupled to the outer race **206** via press-fit, welding, mechanical fasteners, etc. The inner race surface **252** defines an inner perimeter or diameter of the race coupling section **240**, and surrounds a bore defined through the race coupling section **240**. The race coupling section **240** also has a first race end **254** opposite a second race end **256**. The first race end **254** is defined at the first housing end **246** and the second race end **256** is coupled to the frangible section **244**. With additional reference to FIG. 5, the race coupling section **240** generally extends for a first distance **D1** from the first housing end **246** to the frangible section **244** along the axis of rotation **140**. In one example, the first distance **D1** is about 0.8 inches (in.) to about 1.2 inches (in.). The race coupling section **240** has a first thickness **T1** defined between the outer race surface **250** and the inner race surface **252**. The first thickness **T1** is uniform over the race coupling section **240**. In one example, the first thickness **T1** is about 0.1 inches (in.) to about 0.5 inches (in.).

With reference back to FIG. 3, the static coupling section **242** is frustoconical, and is sized to extend from the frangible section **244** to couple to the flange **212** associated with the static structure **210** (FIG. 2). In this example, the static coupling section **242** tapers from the second housing end **248** to the frangible section **244**. With reference to FIG. 4, the static coupling section **242** includes an outer static surface **260** opposite an inner static surface **262**. The outer static surface **260** forms an outer surface or outer perimeter of the static coupling section **242**, and the inner static surface **262** faces the shaft **130**. The inner static surface **262** defines an inner perimeter or diameter of the static coupling section **242**, and surrounds a bore defined through the static coupling section **242**. The static coupling section **242** also has a first static end **264** opposite a second static end **266**. The first static end **264** is coupled to the frangible section **244**, and the second static end **266** is defined at the second housing end **248**. The second static end **266** includes a flange **268**, which extends radially about the perimeter or circumference of the static coupling section **242** at the second static end **266**. The flange **268** may include one or more bores **269**, which receive the one or more bolts **216** (FIG. 1) to couple the bearing housing **208** to the static structure **210**. With additional reference to FIG. 5, the static coupling section **242** generally extends for a second distance **D2** from the frangible section **244** to the second housing end **248** along the axis of rotation **140**. In one example, the second distance **D2** is about 3.0 inches (in.) to about 3.5 inches (in.). The

second distance D2 is different and greater than the distance D1. The static coupling section 242 has a second thickness T2 defined between the outer static surface 260 and the inner static surface 262. The second thickness T2 is uniform over the static coupling section 242. The second thickness T2 is about the same as the thickness T1.

With reference back to FIGS. 3 and 4, the frangible section 244 is defined between the race coupling section 240 and the static coupling section 242. The frangible section 244 is substantially L-shaped, and includes a first frangible portion 270 and a second frangible portion 272. With reference to FIG. 5, the first frangible portion 270 extends along an axis A1, which is substantially perpendicular to the axis of rotation 140. Thus, generally, the first frangible portion 270 extends radially from and relative to the race coupling section 240. The first frangible portion 270 includes a first outer surface 274 opposite a first inner surface 276. The first outer surface 274 defines an outer surface or outer perimeter of the first frangible portion 270, and the first inner surface 276 defines an inner surface or perimeter of the first frangible portion 270. The first frangible portion 270 also has a first portion end 278 opposite a second portion end 280. The first portion end 278 is coupled to the second race end 256 and the second portion end 280 is coupled to the second frangible portion 272. The first frangible portion 270 generally extends for a third distance D3 from the second race end 256 to the second frangible portion 272 in a radial direction relative to the axis of rotation 140. In one example, the third distance D3 is about 0.8 inches (in.) to about 1.2 inches (in.). The first frangible portion 270 has a third thickness T3 at the first portion end 278 and the second portion end 280, and has a fourth thickness T4 defined between the first portion end 278 and the second portion end 280. The third thickness T3 is about the same as the first thickness T1 and the second thickness T2. The fourth thickness T4 is different and less than the third thickness T3, and thus, the first thickness T1 and the second thickness T2. In one example, the fourth thickness T4 is about 0.03 inches (in.) to about 0.08 inches (in.). The fourth thickness T4 defines a first area of reduced thickness 282. In this example, the first area of reduced thickness 282 extends for a first length L1 about the perimeter of the bearing housing 208. In one example, the first length L1 is about 0.2 inches (in.) to about 0.5 inches (in.).

In this example, the first area of reduced thickness 282 is defined by a first concave groove 284 recessed into the first outer surface 274 and an opposite second concave groove 286 recessed into the first inner surface 276. The opposed grooves 284, 286 reduce the thickness of the first frangible portion 270 to the fourth thickness T4 between the first portion end 278 and the second portion end 280. By reducing the thickness of the first frangible portion 270 to the fourth thickness T4, the first frangible portion 270 is configured to release, break, or fracture along the first area of reduced thickness 282 to inhibit the transfer of a moment load that is greater than a predetermined moment threshold. In one example, the moment threshold is about 70,000 pound-force inch (lbf-in.) to about 90,000 pound-force inch (lbf-in.). By releasing, breaking or fracturing along the first area of reduced thickness 282 upon receipt of a moment load greater than the moment threshold, the first frangible portion 270 isolates the moment load from the static structure 210 (FIG. 2). This inhibits the static structure 210 from absorbing the moment load.

The second frangible portion 272 extends along an axis A2, which is substantially parallel to the axis of rotation 140. Thus, generally, the second frangible portion 272 extends

axially relative to the race coupling section 240. The second frangible portion 272 includes a second outer surface 290 opposite a second inner surface 292. The second outer surface 290 defines an outer surface or outer perimeter of the second frangible portion 272, and the second inner surface 292 defines an inner surface or perimeter of the second frangible portion 272. The second frangible portion 272 also has a third portion end 294 opposite a fourth portion end 296. The third portion end 294 is coupled to the first frangible portion 270 and the fourth portion end 296 is coupled to the static coupling section 242. The second frangible portion 272 generally extends for a fourth distance D4 from the first frangible portion 270 to the static coupling section 242 along the axis of rotation 140. In one example, the fourth distance D4 is about 0.8 inches (in.) to about 1.2 inches (in.). The second frangible portion 272 has a fifth thickness T5 at the third portion end 294 and the fourth portion end 296, and has a sixth thickness T6 defined between the third portion end 294 and the fourth portion end 296. The fifth thickness T5 is about the same as the first thickness T1, the second thickness T2 and the third thickness T3. The sixth thickness T6 is different and less than the first thickness T1, the second thickness T2 and the third thickness T3. In one example, the sixth thickness T6 is about the same as the fourth thickness T4. The sixth thickness T6 defines a second area of reduced thickness 298. In this example, the second area of reduced thickness 298 extends for a second length L2 about the perimeter of the bearing housing 208. In one example, the second length L2 is about the same as the first length L1. The first area of reduced thickness 282 is spaced a distance apart from the second area of reduced thickness 298.

In this example, the second area of reduced thickness 298 is defined by a third concave groove 300 recessed into the second outer surface 290 and an opposite fourth concave groove 302 recessed into the second inner surface 292. The opposed grooves 300, 302 reduce the thickness of the second frangible portion 272 to the sixth thickness T6 between the third portion end 294 and the fourth portion end 296. By reducing the thickness of the second frangible portion 272 to the sixth thickness T6, the second frangible portion 272 is configured to release, break or fracture along the second area of reduced thickness 298 to inhibit the transfer of a shear or radial load that is greater than a predetermined shear or radial threshold. In one example, the radial threshold is about 45,000 pound-force (lbf) to about 60,000 pound-force (lbf). By releasing, breaking or fracturing along the second area of reduced thickness 298 upon receipt of a radial load greater than the radial threshold, the second frangible portion 272 isolates the radial load from the static structure 210 (FIG. 2). This inhibits the static structure 210 from absorbing the radial load.

In this example, the grooves 284, 286, 300, 302 are defined to extend about the circumference of the bearing housing 208 on the respective one of the surfaces 274, 276, 290, 292, however, in other examples, one of the grooves 284, 286, 300, 302 may be defined to a greater depth on one of the surfaces 274, 276, 290, 292 such that the first frangible portion 270 and the second frangible portion 272 have the respective one of the fourth thickness T4 and the sixth thickness T6 with a single groove. In addition, while the areas of reduced thickness 282, 298 are described as areas, the grooves 284, 286, 300, 302 may be defined to result in a line of reduced thickness, if desired.

In addition, it should be noted that while the first frangible portion 270 and the second frangible portion 272 of the bearing housing 208 are shown in FIGS. 1-5 as comprising

continuous grooves **284**, **286**, **300**, **302** defined about the perimeter or circumference, in other embodiments, the first frangible portion **270** and the second frangible portion **272** may be configured differently to release, fracture or break upon receipt of a respective one of a predetermined moment load and radial load. For example, with reference to FIG. 6, a bearing housing **408** is shown for use with the gas turbine engine **100** (FIG. 1). As the bearing housing **408** includes components that are the same or similar to components of the bearing housing **208** discussed with regard to FIGS. 1-5, the same reference numerals will be used to denote the same or similar components. The bearing housing **408** is coupled to the outer race **206**, and to the flange **212** associated with the static structure **210** (FIG. 2). The bearing housing **408** is composed of a metal or metal alloy, and is cast, forged, stamped, machined, additively manufactured, etc. In this example, the bearing housing **408** includes the first coupling section or race coupling section **240**, the second coupling section or static coupling section **242** and a frangible section **444** between the race coupling section **240** and the static coupling section **242**. In this example, the bearing housing **408** is substantially symmetric about the axis of rotation **140**. The race coupling section **240** is at the first housing end **246**, and the static coupling section **242** extends from the frangible section **444** to the second housing end **248**. The second housing end **248** is opposite the first housing end **246**.

The race coupling section **240** is cylindrical, and is sized to be positioned about the outer race **206**, the bearing elements **204** and the inner race **202** (FIG. 2). With reference to FIG. 7, the race coupling section **240** has the first thickness **T1** defined between the outer race surface **250** and the inner race surface **252**. The static coupling section **242** is frustoconical, and is sized to extend from the frangible section **444** to couple to the flange **212** associated with the static structure **210** (FIG. 2). The static coupling section **242** has the second thickness **T2** defined between the outer static surface **260** and the inner static surface **262**.

The frangible section **444** is defined between the race coupling section **240** and the static coupling section **242**. The frangible section **444** is substantially L-shaped, and includes a first frangible portion **470** and a second frangible portion **472**. The first frangible portion **470** extends along the axis **A1**, which is substantially perpendicular to the axis of rotation **140**. Thus, generally, the first frangible portion **470** extends radially from and relative to the race coupling section **240**. The first frangible portion **470** includes a first outer surface **474** opposite a first inner surface **476**. The first outer surface **474** defines an outer surface or outer perimeter of the first frangible portion **470**, and the first inner surface **476** defines an inner surface or perimeter of the first frangible portion **470**. The first frangible portion **470** also has a first portion end **478** opposite a second portion end **480**. The first portion end **478** is coupled to the second race end **256** and the second portion end **480** is coupled to the second frangible portion **472**. The first frangible portion **470** generally extends for the third distance **D3** from the second race end **256** to the second frangible portion **472** in a radial direction relative to the axis of rotation **140**. The first frangible portion **470** has the third thickness **T3** at the first portion end **478** and the second portion end **480**.

In this example, with reference back to FIG. 6, the first frangible portion **470** includes a plurality of first slots **482** defined between the first portion end **478** and the second portion end **480** about the perimeter of the first frangible portion **470**. Each of first slots **482** are substantially oval in shape, and are spaced apart about the perimeter of the first frangible portion **470**. Each of the first slots **482** have a

major axis **M1**, which is substantially perpendicular to the axis of rotation **140**. The first slots **482** are defined through the first outer surface **474** and the first inner surface **476** such that the first slots **482** result in a void or an area of the first frangible portion **470** that is removed. The formation of the first slots **482** about the perimeter of the first frangible portion **470** also results in a plurality of first ligaments **484** being defined about the perimeter of the first frangible portion **470**. A respective first ligament **484** is defined between each of the first slots **482**. Each of the first ligaments **484** is configured to release, break or fracture to inhibit the transfer of the moment load that is greater than the predetermined moment threshold. By releasing, breaking or fracturing along the first ligaments **484** upon receipt of the moment load greater than the moment threshold, the first frangible portion **470** isolates the moment load from the static structure **210** (FIG. 2). This inhibits the static structure **210** from absorbing the moment load.

With reference to FIG. 7, the second frangible portion **472** extends along the axis **A2**, which is substantially parallel to the axis of rotation **140**. Thus, generally, the second frangible portion **472** extends axially relative to the race coupling section **240**. The second frangible portion **472** includes a second outer surface **490** opposite a second inner surface **492**. The second outer surface **490** defines an outer surface or outer perimeter of the second frangible portion **472**, and the second inner surface **492** defines an inner surface or perimeter of the second frangible portion **472**. The second frangible portion **472** also has a third portion end **494** opposite a fourth portion end **496**. The third portion end **494** is coupled to the first frangible portion **470** and the fourth portion end **496** is coupled to the static coupling section **242**. The second frangible portion **472** generally extends for the fourth distance **D4** from the first frangible portion **470** to the static coupling section **242** relative to the axis of rotation **140**. The second frangible portion **472** has the fifth thickness **T5** at the third portion end **494** and the fourth portion end **496**.

In this example, the second frangible portion **472** includes a plurality of second slots **498** defined between the third portion end **494** and the fourth portion end **496** about the perimeter of the second frangible portion **472**. Each of second slots **498** are substantially oval in shape, and are spaced apart about the perimeter of the second frangible portion **472**. Each of the second slots **498** have a second major axis **M2**, which is substantially parallel to the axis of rotation **140**. The second slots **498** are spaced a distance apart from the first slots **482**. The second slots **498** are defined through the second outer surface **490** and the second inner surface **492** such that the second slots **498** result in a void or an area of the second frangible portion **472** that is removed. The formation of the second slots **498** about the perimeter of the second frangible portion **472** also results in a plurality of second ligaments **500** being defined about the perimeter of the second frangible portion **472**. A respective second ligament **500** is defined between each of the second slots **498**. Each of the second ligaments **500** is configured to release, break or fracture to inhibit the transfer of the shear or radial load that is greater than the predetermined radial threshold. By releasing, breaking or fracturing along the second ligaments **500** upon receipt of the radial load greater than the radial threshold, the second frangible portion **472** isolates the radial load from the static structure **210** (FIG. 2). This inhibits the static structure **210** from absorbing the radial load.

In addition, it should be noted that while the first frangible portion **270** and the second frangible portion **272** of the

11

bearing housing 208 are shown in FIGS. 1-5 as comprising continuous grooves 284, 286, 300, 302 defined about the perimeter or circumference, in other embodiments, the first frangible portion 270 and the second frangible portion 272 may be configured differently to release, fracture or break upon receipt of a respective one of a predetermined moment load and radial load. For example, with reference to FIG. 8, a bearing housing 608 is shown for use with the gas turbine engine 100 (FIG. 1). As the bearing housing 608 includes components that are the same or similar to components of the bearing housing 208 discussed with regard to FIGS. 1-5, the same reference numerals will be used to denote the same or similar components. The bearing housing 608 is coupled to the outer race 206, and to the flange 212 associated with the static structure 210 (FIG. 2). The bearing housing 608 is composed of a metal or metal alloy, and is cast, forged, stamped, machined, additively manufactured, etc. In this example, the bearing housing 608 includes the first coupling section or race coupling section 240, the second coupling section or static coupling section 242 and a frangible section 644 between the race coupling section 240 and the static coupling section 242. In this example, the bearing housing 608 is substantially symmetric about the axis of rotation 140. The race coupling section 240 is at the first housing end 246, and the static coupling section 242 extends from the frangible section 644 to the second housing end 248. The second housing end 248 is opposite the first housing end 246.

The race coupling section 240 is cylindrical, and is sized to be positioned about the outer race 206, the bearing elements 204 and the inner race 202 (FIG. 2). With reference to FIG. 9, the race coupling section 240 has the first thickness T1 defined between the outer race surface 250 and the inner race surface 252. The static coupling section 242 is frustoconical, and is sized to extend from the frangible section 444 to couple to the flange 212 associated with the static structure 210 (FIG. 2). The static coupling section 242 has the second thickness T2 defined between the outer static surface 260 and the inner static surface 262.

The frangible section 644 is defined between the race coupling section 240 and the static coupling section 242. The frangible section 644 is substantially L-shaped, and includes a first frangible portion 670 and a second frangible portion 672. The first frangible portion 670 extends along the axis A1, which is substantially perpendicular to the axis of rotation 140. Thus, generally, the first frangible portion 670 extends radially from and relative to the race coupling section 240. The first frangible portion 670 includes a first outer surface 674 opposite a first inner surface 676. The first outer surface 674 defines an outer surface or outer perimeter of the first frangible portion 670, and the first inner surface 676 defines an inner surface or perimeter of the first frangible portion 670. The first frangible portion 670 also has a first portion end 678 opposite a second portion end 680. The first portion end 678 is coupled to the second race end 256 and the second portion end 680 is coupled to the second frangible portion 672. The first frangible portion 670 generally extends for the third distance D3 from the second race end 256 to the second frangible portion 672 in a radial direction relative to the axis of rotation 140. The first frangible portion 670 has the third thickness T3 at the first portion end 678 and the second portion end 680.

In this example, with reference back to FIG. 8, the first frangible portion 670 includes a plurality of first slots 682 defined between the first portion end 678 and the second portion end 680 about the perimeter of the first frangible portion 670. Each of first slots 682 are substantially oval in shape, and are spaced apart about the perimeter of the first

12

frangible portion 470. Each of the first slots 682 have a major axis M1, which is substantially perpendicular to the axis of rotation 140. The first slots 682 are defined through the first outer surface 674 and the first inner surface 676 such that the first slots 682 result in a void or an area of the first frangible portion 670 that is removed. The formation of the first slots 682 about the perimeter of the first frangible portion 670 also results in a plurality of first ligaments 684 being defined about the perimeter of the first frangible portion 670. A respective first ligament 684 is defined between each of the first slots 682. Each of the first ligaments 684 is configured to release, break or fracture to inhibit the transfer of the moment load that is greater than the predetermined moment threshold. By releasing, breaking, or fracturing along the first ligaments 684 of the first frangible portion 670 upon receipt of the moment load greater than the moment threshold, the first frangible portion 670 isolates the moment load from the static structure 210 (FIG. 2). This inhibits the static structure 210 from absorbing the moment load.

With reference to FIG. 9, the second frangible portion 672 extends along the axis A2, which is substantially parallel to the axis of rotation 140. Thus, generally, the second frangible portion 672 extends axially relative to the race coupling section 240. The second frangible portion 672 includes a second outer surface 690 opposite a second inner surface 692. The second outer surface 690 defines an outer surface or outer perimeter of the second frangible portion 672, and the second inner surface 692 defines an inner surface or perimeter of the second frangible portion 672. The second frangible portion 672 also has a third portion end 694 opposite a fourth portion end 696. The third portion end 694 is coupled to the first frangible portion 670 and the fourth portion end 696 is coupled to the static coupling section 242. The second frangible portion 672 generally extends for the fourth distance D4 from the first frangible portion 670 to the static coupling section 242 relative to the axis of rotation 140. The second frangible portion 672 has the fifth thickness T5 at the third portion end 694 and the fourth portion end 696.

In this example, the second frangible portion 672 includes a plurality of second slots 698 defined between the third portion end 694 and the fourth portion end 696 about the perimeter of the second frangible portion 672. Each of second slots 698 are substantially oval in shape, and are spaced apart about the perimeter of the second frangible portion 672. Each of the second slots 698 have a second major axis M2, which is substantially perpendicular to the axis of rotation 140. The second slots 698 are spaced a distance apart from the first slots 682. The second slots 698 are defined through the second outer surface 690 and the second inner surface 692 such that the second slots 698 result in a void or an area of the second frangible portion 672 that is removed. The formation of the second slots 698 about the perimeter of the second frangible portion 672 also results in a plurality of second ligaments 700 being defined about the perimeter of the second frangible portion 672. A respective second ligament 700 is defined between each of the second slots 598. Each of the second ligaments 700 is configured to release, break or fracture to inhibit the transfer of the shear or radial load that is greater than the predetermined radial threshold. By releasing, breaking or fracturing along the second ligaments 700 upon receipt of the radial load greater than the radial threshold, the second frangible portion 672 isolates the radial load from the static structure 210 (FIG. 2). This inhibits the static structure 210 from absorbing the radial load.

13

It should be noted that while the bearing housing **208**, **408**, **608** is illustrated herein as including the grooves **284**, **286**, **300**, **302**, the slots **482**, **498** and the slots **682**, **698**, respectively, a bearing housing may include any combination of the grooves **284**, **286**, **300**, **302**, the slots **482**, **498** and the slots **682**, **698** to enable the bearing housing to isolate moment and radial loads from the static structure **210** (FIG. 2).

In one example, with reference to FIGS. 2-9, the inner race **202**, the bearing elements **204** and the outer race **206** are coupled to the shaft **130**. With the bearing housing **208**, **408**, **608** formed, the bearing housing **208**, **408**, **608** is coupled to the outer race **206** such that the outer race **206** is received within and coupled to the race coupling section **240**. With the outer race **206** coupled to the bearing housing **208**, **408**, **608**, the flange **268** is coupled to the static structure **210** via the bolts **216**. With the bearing housing **208**, **408**, **608** installed in the gas turbine engine **100** (FIG. 1), as the gas turbine engine **100** operates, the shaft **130** rotates.

In the instance of a moment load being imparted to the bearing assembly **200** that is greater than the moment threshold, the first frangible portion **270**, **470**, **670** releases, breaks or fractures to absorb the load and isolate the static structure **210** from the moment load. In the example of the first frangible portion **270**, the first frangible portion **270** releases, breaks or fractures along the grooves **284**, **286** upon receipt of the moment load that is greater than the moment threshold. In the example of the first frangible portion **470**, **670**, the first frangible portion **470**, **670** releases, breaks or fractures along the first ligaments **484**, **684** upon receipt of the moment load that is greater than the moment threshold.

In the instance of a shear or radial load being imparted to the bearing assembly **200** that is greater than the radial threshold, the second frangible portion **272**, **472**, **672** releases, breaks or fractures to absorb the load and isolate the static structure **210** from the radial load. In the example of the second frangible portion **272**, the second frangible portion **272** releases, breaks or fractures along the grooves **300**, **302** upon receipt of the radial load that is greater than the radial threshold. In the example of the second frangible portion **472**, **672**, the second frangible portion **472**, **672** releases, breaks or fractures along the second ligaments **500**, **700** upon receipt of the radial load that is greater than the radial threshold.

Thus, the bearing assembly **200** including the bearing housing **208**, **408**, **608** provides load isolation for the gas turbine engine **100** (FIG. 1), which inhibits the transfer of moment and radial loads to the static structure **210** through the bearing assembly **200**. Each of the first frangible portion **270**, **470**, **670** and the second frangible portion **272**, **472**, **672** may release, break or fracture independently upon receipt of the respective moment load or radial load that exceeds the respective moment threshold or radial threshold. This inhibits the static structure **210** (FIG. 2) from absorbing the respective moment load or radial load, which inhibits damage to the static structure **210** from moment or radial loads.

In this document, relational terms such as first and second, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. Numerical ordinals such as "first," "second," "third," etc. simply denote different singles of a plurality and do not imply any order or sequence unless specifically defined by the claim language. The sequence of the text in any of the claims does not imply that process steps must be performed in a temporal or logical order according to such sequence unless it is specifically defined by the

14

language of the claim. The process steps may be interchanged in any order without departing from the scope of the invention as long as such an interchange does not contradict the claim language and is not logically nonsensical.

While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

What is claimed is:

1. A bearing assembly, comprising:

an inner race configured to receive a rotating component having an axis of rotation;

at least one bearing element coupled to the inner race, the at least one bearing element configured to rotate with the inner race along the axis of rotation;

an outer race coupled to the at least one bearing element; and

a bearing housing coupled to the outer race and configured to be coupled to a static structure, the bearing housing including a first frangible portion configured to release upon receipt of a moment load greater than a moment threshold and a second frangible portion configured to release upon receipt of a radial load greater than a radial threshold.

2. The bearing assembly of claim 1, wherein the first frangible portion is defined on the bearing housing along an axis substantially perpendicular to the axis of rotation.

3. The bearing assembly of claim 1, wherein the second frangible portion is defined on the bearing housing along an axis substantially parallel to the axis of rotation.

4. The bearing assembly of claim 1, wherein the first frangible portion includes at least one first groove that defines a reduced thickness.

5. The bearing assembly of claim 4, wherein the second frangible portion includes at least one second groove that defines a second reduced thickness.

6. The bearing assembly of claim 4, wherein the at least one first groove is continuous about a perimeter of the first frangible portion.

7. The bearing assembly of claim 4, wherein the at least one first groove comprises a pair of opposed grooves that cooperate to define the reduced thickness.

8. The bearing assembly of claim 1, wherein the first frangible portion includes a plurality of first slots spaced apart about a perimeter of the first frangible portion.

9. The bearing assembly of claim 8, wherein the second frangible portion includes a plurality of second slots spaced apart about a second perimeter of the second frangible portion.

10. The bearing assembly of claim 9, wherein each second slot extends along a major axis, and the major axis is substantially parallel to the axis of rotation.

11. The bearing assembly of claim 8, wherein each first slot extends along a major axis, and the major axis is substantially perpendicular to the axis of rotation.

12. The bearing assembly of claim 1, wherein the bearing housing includes a first coupling section to be coupled to the

15

outer race, a second coupling section configured to be coupled to the static structure, and the first frangible portion and the second frangible portion are defined between the first coupling section and the second coupling section.

13. The bearing assembly of claim 12, wherein the first frangible portion extends radially outward from the first coupling section, and the second frangible portion extends axially from the first frangible portion relative to the axis of rotation.

14. The bearing assembly of claim 13, wherein the second frangible portion is coupled to a first end of the second coupling section, and the second coupling section defines a flange at a second end, with the second end opposite the first end.

15. The bearing assembly of claim 1, wherein the rotating component is a shaft associated with a gas turbine engine.

16. A bearing assembly for a gas turbine engine, comprising:

an inner race configured to receive a rotating component having an axis of rotation;

at least one bearing element coupled to the inner race, the at least one bearing element configured to rotate with the inner race along the axis of rotation;

an outer race coupled to the at least one bearing element; and

a bearing housing coupled to the outer race and configured to be coupled to a static structure, the bearing housing including a first frangible portion and a second

16

frangible portion, the first frangible portion extending along a first axis substantially perpendicular to the axis of rotation, the first frangible portion configured to release upon receipt of a moment load greater than a moment threshold, the second frangible portion extending along a second axis substantially parallel to the axis of rotation and the second frangible portion is configured to release upon receipt of a radial load greater than a radial threshold.

17. The bearing assembly of claim 16, wherein the first frangible portion includes a pair of opposed grooves that cooperate to define a reduced thickness and each groove of the pair of opposed grooves is continuous about a perimeter of the first frangible portion.

18. The bearing assembly of claim 16, wherein the first frangible portion includes a plurality of first slots spaced apart about a perimeter of the first frangible portion and the second frangible portion includes a plurality of second slots spaced apart about a second perimeter of the second frangible portion.

19. The bearing assembly of claim 18, wherein each second slot extends along a major axis, the major axis is substantially parallel to the axis of rotation.

20. The bearing assembly of claim 18, wherein each first slot extends along a major axis, and the major axis is substantially perpendicular to the axis of rotation.

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